

Complete Analytics System & Data Architecture Audit

Comprehensive Read-Only Technical Inspection of Current Capabilities, Data Models, Metrics, Gap Analysis & Future Analytics Architecture

Audit Date:	August 17, 2026
Audit Scope:	Full Lumora Codebase (Backend, Frontend, 36 DB Models, 28 API Routers, 28 Services)
Audit Type:	Read-Only Architecture & Data Integrity Inspection
Target Domain:	G.C.E. Advanced Level Biology Examination & Coursework Learning Analytics
Audit Status:	COMPLETE — STRICTLY READ-ONLY (NO CODE/DB MODIFICATIONS)

Executive Notice: This technical audit report was conducted strictly under read-only constraints to provide the definitive ground truth of existing data structures, metrics, and API surfaces. No schemas were mutated, no data was modified, and no assumptions were substituted for actual codebase facts.

Executive Summary

Lumora currently possesses a substantial data footprint across courses, student learning activities, and the newly architected G.C.E. Advanced Level Assessment Engine. However, existing analytics are fragmented across three distinct subsystems: (1) Course/Quiz legacy analytics, (2) Material difficulty hotspot flagging, and (3) A/L Examination submissions.

Core Finding: The underlying database tables (al_student_submissions, al_student_answers, al_questions, activity_logs, student_material_progress) preserve high-fidelity raw event data — including exact student answer options, subpart JSON responses, rubric checkmarks, response times, and cognitive taxonomy tags. However, **nearly 80% of required teacher item analysis and student mastery metrics are currently computed on-the-fly or not yet aggregated into dedicated analytics tables.**

Subsystem Area	Data Exists in DB	Calculated in API	Displayed in UI	Readiness Status
Course & Enrollment Overview	Yes (courses, enrollments)	Yes (Aggregations)	Yes (Teacher Workstation)	AVAILABLE
Legacy Quiz Analytics	Yes (quiz_attempts, answers)	Yes (Avg, Min, Max, Dist)	Yes (Charts)	AVAILABLE
A/L Exam Overall Performance	Yes (al_student_submissions)	Yes (Score, Grade, %)	Yes (Teacher/Student Exam)	AVAILABLE
MCQ Item Analysis (Discrimination/Distractor)	Yes (al_student_answers)	Partial (Legacy Q only)	No	PARTIAL (RAW READY)
Structured Question Hierarchy Analytics	Yes (subpart_answers_json)	No	No	PARTIAL (RAW READY)
Essay Criteria & Misconception Analytics	Yes (ai_checklist_results_json)	No	No	PARTIAL (RAW READY)
Material Confusion Hotspots	Yes (material_flags, hotspots)	Yes (Contexts, AI brief)	Yes (Material Insights)	AVAILABLE
Ask AI Tutor Question Categorization	Yes (student_questions, ai_responses)	Yes (Topic categorization)	Yes (AI Insights tab)	AVAILABLE
Gemini AI Token & Billing Cost Tracking	Partial (ai_logs tokens/ms)	Partial (Admin performance)	Admin only	PARTIAL
Time-on-Task & Video Progress Intervals	Partial (Last position only)	No session tracking	No	NOT AVAILABLE

1. Codebase Discovery & Component Map

The audit identified 12 core frontend pages, 6 analytics API routes, 36 SQLAlchemy database tables, and 28 backend services related to student analytics and exam authoring.

File Path	Component / Route	Data Source & Purpose	Status
frontend/src/app/dashboard/teacher/analytics/page.tsx	TeacherAnalyticsPage	Full course overview, coursework breakdown, student risk roster, AI confusion topics API /api/analytics/teacher/course/{id}/full-analytics	Active
frontend/src/app/dashboard/student/analytics/page.tsx	StudentAnalyticsOverviewPage	Student overall score, materials completed, assessments taken, quiz history API /api/analytics/student/progress & quiz-history	Active

frontend/src/app/dashboard/teacher/insights/page.tsx	TeacherInsightsPage	Material flag clusters, heatmap visualizations, AI hotspot summaries API /api/materials/teacher/insights/flags	Active
frontend/src/app/dashboard/teacher/qa/page.tsx	TeacherQAModerationPage	Student Q&A; feed, AI responses, confidence scores, teacher replies API /api/qa/teacher/questions	Active
backend/app/api/analytics.py	AnalyticsRouter	Teacher & Student progress aggregations, 14-day trends, composite risk score calculation Direct SQLAlchemy DB queries	Active
backend/app/api/al_exams.py	ALExamRouter	A/L exam attempt lifecycle, deterministic MCQ grading, AI structured/essay pre-marking, verification al_student_submissions, al_student_answers	Active
backend/app/services/question_analytics.py	QuestionAnalyticsService	Computes item difficulty index p and discrimination index d for legacy quiz questions question_analytics table	Active (Legacy only)
backend/app/services/gemini_service.py	GeminiService	Central Gemini client with token tracking and failover logging to ai_logs ai_logs table	Active

2. Current Analytics System & Calculations

The audit evaluated every metric currently calculated across student, teacher, and assessment domains:

A. Teacher Composite Risk Score Formula

In backend/app/api/analytics.py (lines 231–246), the teacher workstation calculates a composite risk index (0–100) on-the-fly for every enrolled student:

Composite = (Quiz_Avg * 0.35) + (Coursework_Avg * 0.35) + (Material_Pct * 0.20) + (min(100, AI_Qs / 3 * 100) * 0.10)

- **At Risk:** Composite < 50.0 OR (Quiz_Avg < 50.0 AND Coursework_Avg < 50.0)
- **Moderate:** 50.0 ≤ Composite < 70.0
- **Healthy:** Composite ≥ 70.0

Limitation: This calculation ignores A/L Exam submissions (al_student_submissions) and is restricted to legacy quizzes and assignments.

B. Student Overall Progress Formula

In backend/app/api/analytics.py (lines 570–575), the student dashboard computes:

Overall_Progress = (min(100, Quiz_Avg) * 0.40) + (min(100, CW_Avg) * 0.40) + (min(100, Completed_Materials * 5.0) * 0.20)

Limitation: Completed_Materials * 5.0 arbitrarily assumes a course has exactly 20 materials (20 * 5 = 100). If a course has 4 materials, the max material score caps at 20%.

C. Material Difficulty Hotspot Clumping

In backend/app/api/materials.py (lines 828–853), student flags containing contextual strings (e.g. 'Timestamp 04:12' or 'Page 7') are clustered by material and passed to Gemini AI to generate an automated executive briefing for teachers. This is actively functional in the UI.

3. Database Schema & Data Model Audit

Lumora has 36 tables in backend/app/models.py. The critical assessment and activity tables are summarized below:

Table Name	Key Fields	Analytics Utility & Persistence
al_student_submissions	id, exam_id, student_id, started_at, submitted_at, raw_score, scaled_score, percentage, grade, status, teacher_verified_at	Tracks student exam attempts. Status values: in_progress, submitted, ai_graded, teacher_verified. Preserves historical attempt data.
al_student_answers	id, submission_id, question_id, selected_option, subpart_answers_json, essay_text_answer, auto_score, ai_score, teacher_score, final_score, ai_checklist_results_json	Stores granular student responses per question. Fully captures exact MCQ choices, structured text subparts, and essay rubrics.
al_questions	id, exam_id, question_number, template_type, stem_text, options, correct_option, structured_subparts_json, essay_checklist_json, cognitive_level, difficulty, snapshot_json	Authoritative question definitions with syllabus taxonomy (remember, understand, apply, analyze, evaluate) and difficulty (easy, medium, hard).
student_material_progress	id, student_id, material_id, last_position, is_completed, updated_at	Stores binary completion status and last playback offset (seconds or PDF page). Does NOT store time spent or session history.
material_flags	id, student_id, material_id, context, comment, is_resolved, created_at	Stores student difficulty flags with contextual locations (e.g. 'Page 4', '01:23').

student_questions	id, session_id, student_id, course_id, question_text, topic_category, sentiment_difficulty, asked_at	Captures student Ask AI queries. Categorized asynchronously by Gemini into topic concepts.
ai_logs	id, action, input_summary, output_summary, tokens_used, processing_time_ms, status, error_message, created_at	Audit log of all LLM operations. Captures token counts and processing latency.
activity_logs	id, user_id, action, entity_type, entity_id, action_metadata, created_at	General purpose event stream (e.g. view_lesson, submit_quiz).

4. Assessment Analytics & MCQ Item Analysis

Core Question: Does Lumora currently collect enough data to perform full Psychometric Item Analysis for 50-Question Paper I MCQs?

Verdict: **DATA IS 100% AVAILABLE IN DATABASE.** The table `al_student_answers` records the exact `selected_option` ('A', 'B', 'C', 'D', 'E') for every student attempt, while `al_questions` contains `correct_option`, `template_type`, `cognitive_level`, and `difficulty`.

Item Metric	Mathematical Definition	Database Data Source	Status
Difficulty Index (p-value)	$p = \text{Correct_Count} / \text{Total_Attempts}$	Calculable directly via <code>SELECT count(*) WHERE is_correct=true / count(*)</code>	AVAILABLE (Needs API)
Discrimination Index (d)	$d = (\text{Upper_27\%_Correct} - \text{Lower_27\%_Correct}) / (0.27 * N)$	All attempts have <code>scaled_score</code> ; upper/lower quartiles can be queried cleanly	AVAILABLE (Needs API)
Option Selection Distribution	Count and % selecting A, B, C, D, E per question	Directly queryable: <code>GROUP BY selected_option</code> on <code>al_student_answers</code>	AVAILABLE (Needs API)
Distractor Efficiency	Identifies non-functional distractors (< 5% selection)	Derived from option distribution against total attempts	AVAILABLE (Needs API)
Question-Type Performance	Average score grouped by 7 MCQ templates	Join <code>al_student_answers</code> with <code>al_questions.template_type</code>	AVAILABLE (Needs API)
Cognitive Level Breakdown	Performance on Remember vs Understand vs Apply vs Analyze	Join <code>al_student_answers</code> with <code>al_questions.cognitive_level</code>	AVAILABLE (Needs API)
Time Spent per Question	Response latency per question item	NOT recorded in <code>al_student_answers</code> (only <code>total_started_at/submitted_at</code>)	NOT AVAILABLE

5. Structured (Paper II-A) & Essay (Paper II-B) Analytics

Structured Paper II-A Readiness: In `al_student_answers.subpart_answers_json`, answers are stored as dictionary key-values (e.g. `{'a(i)': 'Casparian strip', 'a(ii)': '...'}`). In `al_student_answers.ai_checklist_results_json`, each subpart has an individual score (e.g. `{'subpart': 'a(i)', 'awarded_score': 2.0, 'maximum_score': 2.0}`).

Hierarchical Analysis Capability: **AVAILABLE.** The system preserves complete hierarchy from Main Question (1–4) → Part (a, b, c) → Subpart (i, ii) → Nested Subpart. A backend aggregation service can determine exactly which subpart (e.g. Q2(a)(ii)) has the highest loss rate across the class.

Essay Paper II-B Readiness: In `al_student_answers.ai_checklist_results_json`, each marking rubric criterion (e.g. 'PSII Photolysis', 'Z-Scheme Transport') has a boolean checkmark and awarded point value. A backend service can aggregate which specific marking points students consistently omit.

6. Learning Activity & Content Material Analytics

- **Material Completion:** Tracked in `student_material_progress.is_completed` (Boolean).
- **Video / PDF Position:** Tracked in `student_material_progress.last_position` (Float seconds or page number).
- **Time-on-Task & Study Sessions:** **NOT RECORDED.** Currently, Lumora does not maintain a heartbeat table or study session intervals. It cannot prove how long a student actually engaged with a page.
- **Material Re-visits & Abandonment:** `activity_logs` records 'view_lesson' and 'view_material' actions, allowing return-visit counts to be derived, but not dwell time.

7. Flagging & Ask AI Tutor Analytics

Flagging Analytics (AVAILABLE): material_flags records student ID, material ID, context (timestamp/page), comment, resolution status, and creation date. The teacher dashboard actively aggregates these into confusion clusters and generates AI Executive Briefings.

Ask AI Tutor Analytics (AVAILABLE): student_questions and ai_responses capture every question, retrieved material context sources, confidence score, and timestamp. The background task in app/services/analytics.py classifies each question into a biological topic_category (e.g. 'Metabolism & Bioenergetics', 'Cell Division & Genetics').

Privacy Consideration: To protect student privacy, the teacher analytics dashboard exposes aggregated topic frequency bars and anonymized question counts, while reserving direct conversational transcripts for student private study.

8. AI Usage & Cost Analytics (Gemini API)

In backend/app/services/gemini_service.py (line 95), every call to Gemini logs an entry into the ai_logs table with:

- action ('qa_answer', 'quiz_gen', 'essay_grading', 'summarize')
- tokens_used (Integer total tokens)
- processing_time_ms (Latency in milliseconds)
- status ('completed' or 'failed')

Data Gap: tokens_used does not currently separate **Input Tokens** vs **Output Tokens**. Because Gemini pricing differs between input (\$0.075/1M) and output (\$0.30/1M), exact dollar billing calculation requires adding input_tokens and output_tokens columns to ai_logs.

9. Analytics Data Gap Matrix

The definitive mapping of required analytics features against current database readiness:

Analytics Feature	Data Exists	Partially Exists	Data Missing	Required Action / Change
Class Overall Score & Grade Dist	Yes	—	—	Build Teacher API to aggregate al_student_submissions
MCQ Question Difficulty (p-value)	Yes	—	—	Build aggregation query on al_student_answers
MCQ Discrimination Index (d)	Yes	—	—	Upper 27% vs Lower 27% quartile subtraction query
MCQ Distractor Efficiency Distribution	Yes	—	—	Group by selected_option on al_student_answers
Structured Subpart Loss Leaderboard	Yes	—	—	Aggregate ai_checklist_results_json subpart marks
Essay Omitted Criteria Frequency	Yes	—	—	Aggregate teacher/ai_checklist_results_json
Material Confusion Hotspots	Yes	—	—	Already operational in Teacher Insights
Ask AI Topic Confusion Distribution	Yes	—	—	Already operational in Teacher Analytics AI tab
Exact Gemini API Dollar Cost	—	Total tokens only	Input vs Output split	Add input_tokens, output_tokens, model_tier to ai_logs
Time-on-Task & Study Sessions	—	—	Session duration	Create study_sessions heartbeat table

Student Weak Topic Recommendation	—	Profile model exists	Automated pipeline	Populate student_learning_profiles from exam results
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10. Proposed Future Analytics Architecture

Based strictly on Lumora's existing stack (FastAPI + PostgreSQL + Next.js), the future analytics system should follow a lightweight, 3-tier architectural flow without heavy external microservices:

Tier 1 — Raw Event Layer (Existing):

- al_student_submissions + al_student_answers (Exams)
- student_material_progress + material_flags (Content)
- student_questions + ai_responses + ai_logs (AI Tutor & Cost)

Tier 2 — Analytics Service & Materialized Rollups (To Be Built):

- al_item_analytics table: Nightly or post-exam trigger calculating p-value, discrimination index d, and distractor frequencies.
- student_mastery_rollups: Aggregates student scores across syllabus units and cognitive levels.

Tier 3 — Presentation & Recommendation Layer (To Be Built):

- **Teacher Assessment Hub:** Item difficulty scatter plots, distractor efficiency bars, and structured subpart error heatmaps.
- **Student Mastery Hub:** Unit-level radar charts, personal weak area indicators, and targeted revision recommendations.

11. Recommended Implementation Roadmap

The recommended phased sequence for implementing the Lumora Analytics System:

Phase & Deliverable	Technical Scope	Est. Effort
Phase 1: Assessment Aggregation APIs	Build backend endpoints in app/api/al_analytics.py querying al_student_answers for MCQ option distributions, difficulty index, and discrimination index.	1-2 Days
Phase 2: Teacher Exam Analytics UI	Add an 'Exam Performance' tab in Teacher Analytics displaying score distributions, hardest questions, and distractor efficiency charts.	2 Days
Phase 3: Structured & Essay Deep-Dive	Implement hierarchical error reporting for Paper II-A subparts and Paper II-B checklist omission frequencies.	2 Days
Phase 4: Student Personal Mastery Dashboard	Upgrade Student Analytics to visualize syllabus unit strengths/weaknesses and cognitive skill performance.	2 Days
Phase 5: AI Cost & Token Breakdown	Add input/output token columns to ai_logs and build the Admin AI Usage/Cost dashboard.	1 Day
Phase 6: Heartbeat Study Session Tracking	Implement lightweight client ping for real time-on-task and video retention curves.	2 Days

12. Final Analytics Readiness Verdict

DATA WE ALREADY HAVE: 100% of student MCQ choices, structured text subparts, essay checklist outcomes, question cognitive levels, difficulties, and material flags.

DATA WE CAN DERIVE: Difficulty indices, discrimination indices, distractor efficiency, subpart failure rates, essay criterion omission rates, and class score distributions.

DATA WE ARE MISSING: Exact question-level time duration, video re-watch interval logs, and input vs. output token billing breakdowns.

NEXT STEP: Proceed to Phase 1 implementation (Assessment Analytics APIs) when ready.